

DISEIL: Demonstration Distillation for Sample-Efficient Imitation Learning

Anonymous submission

Abstract

An expert demonstration is the one input to imitation learning whose cost does not fall when compute is bought or a simulator is downloaded, so under a fixed allowance the only lever left on policy performance is what each demonstration contains. Robot-gated query-efficient interactive imitation learning reduces to one template: score a visited state, compare against a threshold, hand control to the expert at the first crossing. That template decides when to call the expert and fixes the other two decisions, which failure to correct and where the demonstration begins, by default. We present **DISEIL** (**D**emonstration **d**istillation for **S**ample-**E**fficient **I**mitation **L**earning), which takes the flag as given and decides the other two: it partitions a round’s failures into modes over a geometric descriptor and prescribes the configuration of a demonstration not yet collected, checked for feasibility before any expert time is spent. We evaluate on five tasks under two observation modalities, ten task-modality settings in all, at a budget of twenty demonstrations, against the query-efficient DAgger family and a uniform-random allocation control. DISEIL holds the best or joint-best mean held-out success rate in all ten, by a mean of 2.80 points over the strongest baseline in each.

Introduction

Behaviour cloning fits a policy to an expert’s actions on the states the expert visits (Pomerleau 1988; Bain and Sammut 1995), and it fails under covariate shift: the learner’s own actions carry it off the demonstrated support, where errors compound (Shimodaira 2000; Ross, Gordon, and Bagnell 2011). Interactive imitation learning removes the compounding by aggregating corrections on the states the learner visits (Ross, Gordon, and Bagnell 2011; Celemin et al. 2022), and pays for it in expert effort, the cost that binds (Mandlekar et al. 2018; Khazatsky et al. 2024). Performance scales with the coverage of a policy’s demonstrations, while those added within a covered region saturate (Lin et al. 2024), so once an allowance of B demonstrations is fixed, the content of each is all that is left to choose (Figure 1).

The query-efficient members of the DAgger family gate the expert behind a per-state predicate on action discrepancy (Zhang and Cho 2017), predictive uncertainty from dropout (Menda, Driggs-Campbell, and Kochenderfer 2017) or an ensemble (Menda, Driggs-Campbell, and Kochenderfer 2019), novelty and risk (Hoque et al. 2021a), or a diffusion policy’s own per-step denoising loss (Lee, Kang, and Kuo 2025; Chi

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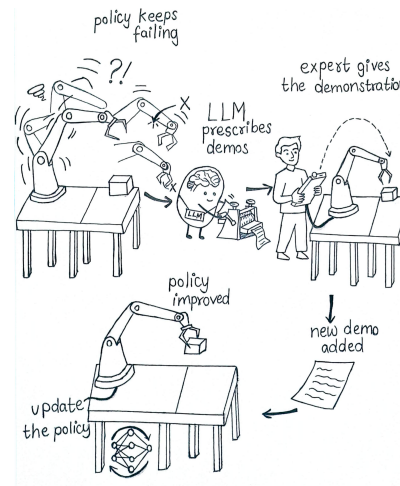


Figure 1: A round of interactive imitation learning ends with a set of failures and one demonstration to spend on it. The gate decides only when to ask. DISEIL decides which failure the call is spent on and what configuration the expert is asked to start from, so the demonstration stands for a mode rather than for the episode that happened to trip the gate first.

et al. 2023). They differ in the scalar and share the template. An acquisition has three parts, when to call the expert, which of the round’s failures to spend the call on, and where the corrective demonstration begins. The template computes the first and takes the other two by default: the failure corrected is whichever rollout tripped the gate first, and the demonstration begins in the state that rollout was already in. Under an unbounded budget the two defaults cost nothing, since every mode is corrected eventually. Under a fixed budget they are the whole problem.

Neither missing decision can be recovered from a scalar attached to one state. It carries no representation in which two failures are the same mistake and a third is a different one, so a gate corrects the recurring mode as often as the recurrence throws it up while a rare mode waits. It also inherits the state that tripped it, frequently one the policy has already ruined, so the expert demonstrates recovery from a corruption instead of avoidance of it. The failures of a round are not independent events but a small set with structure,

and that structure is recoverable from geometry: a descriptor of the configuration at the step where the policy first becomes unreliable partitions them into geometrically distinct groups. Whether those groups are worth allocating over is an empirical question, and the ablations answer it.

Demonstration distillation for sample-efficient imitation learning (DISEIL) takes the flag as given and computes the other two parts of the acquisition. Each round it anchors every failure at the first threshold crossing rather than at the peak, partitions the failures into modes over that descriptor, and targets the near-dominant mode of highest mean peak loss. It then prescribes the configuration of a demonstration not yet collected, verified against a constraint graph of what the task permits. The round’s failure set is distilled to one demonstration, chosen so that what the expert supplies stands for the mode rather than for one episode.

We contribute three things. We separate the acquisition rule into when, which and where, and show the DAgger family fixes the last two by default. We present DISEIL, which attains the best or joint-best mean held-out success rate in all ten settings, by a mean of 2.80 points over the strongest baseline in each. And we locate the advantage in allocation rather than in per-demonstration informativeness: removing the mode structure costs a mean 4.37 points of success while information gain does not fall, the signature of a redundantly spread budget.

Related Work

Interactive imitation learning. Dataset aggregation corrects the covariate shift of behaviour cloning (Shimodaira 2000) by labelling the learner’s own visited states and re-fitting on the aggregate, trading a bound quadratic in the horizon for a linear one (Ross and Bagnell 2010; Ross, Gordon, and Bagnell 2011), a skeleton every method compared here shares. Intervention-weighted methods reweight the aggregate once collected (Mandlekar et al. 2020; Liu et al. 2023b); they change how the round’s data is fitted, not which data it buys.

The query gates. Aggregation asks the expert to label every visited state, so query-efficient variants insert a gate: score each visited state, threshold it, and hand over at the first crossing. The five compared here differ only in the score: a classifier predicting deviation from the expert (Zhang and Cho 2017), the concentration of dropout samples about the expert’s action (Menda, Driggs-Campbell, and Kochenderfer 2017), the variance of a policy ensemble combined with an action discrepancy (Menda, Driggs-Campbell, and Kochenderfer 2019), a novelty term and a task-risk estimate (Hoque et al. 2021a), and the per-step denoising loss of a diffusion policy (Chi et al. 2023), which Diff-Dagger reads as an uncertainty signal (Lee, Kang, and Kuo 2025). DISEIL uses that same quantity to localise a failure and to measure information gain. Human-gated variants leave the decision to the operator (Kelly et al. 2019), and others vary the switch-back rule rather than the score (Hoque et al. 2021b). Each gate answers when to hand over, and neither the which nor the where follows from any of them.

Demonstration selection. Active learning ranks candidates by an acquisition function (Settles 2009), through uncertainty (Houlsby et al. 2011) or coverage, which core-set solves by a greedy k -centre rule (Sener and Savarese 2018) and which BADGE reaches by k -means++ over gradient embeddings (Ash et al. 2020); sub-trajectory retrieval ranks segments of a demonstration corpus (Mommel et al. 2025). These methods rank a pool that already exists. A round’s failure set is such a pool, and the A3 knockout of the Ablations ranks it by uncertainty alone, which is that family’s rule with the coverage term removed. A coverage rule over the failure set is the nearest competitor to the partition and is not run here. What none of them can do is specify a configuration the pool does not contain, which is what the prescription supplies. Dataset distillation compresses a held training set into a small synthetic one (Cazenavette et al. 2022); it runs after collection, whereas demonstration distillation runs before it.

Language models and standard components. Language models have been used in robotics as planners, code generators and reward designers (Ahn et al. 2022; Liang et al. 2023; Ma et al. 2024), and a model given frames and an execution record as structured evidence can name a failure’s cause well enough to drive recovery (Liu, Bahety, and Song 2023; Duan et al. 2025). The same models are unreliable at metric and spatial reasoning from pixels (Chen et al. 2024a; Fu et al. 2024). DISEIL therefore hands the model the geometry and a constraint graph of what the environment permits, and asks only for a cause and a request; no model here emits an action. Proposing to an external solver is established (Liu et al. 2023a), as is returning a checker’s verdict as feedback for re-proposal (Chen et al. 2024b); what differs is the object verified, a demonstration not yet collected rather than a plan to be executed. The remaining machinery is standard and is not claimed: the clustering, the cluster count, the diversity rule and the grid path checker are named with their sources in the Method, at the point each is used.

Problem Formulation

The policy is any function f_θ mapping an observation to an action that exposes a per-step loss at a visited state-action pair,

$$\ell_t^{(i)} = \mathcal{L}(f_\theta, s_t^{(i)}, a_t^{(i)}), \quad (1)$$

where i indexes an episode and $a_t^{(i)}$ is the action the policy itself executed. The requirement stops there. For a diffusion policy $\mathcal{L}$ is the denoising loss (Chi et al. 2023); for a discrete policy it is the negative log-likelihood of the action.

The policy is fitted to an initial set $\mathcal{D}_0$ by behaviour cloning (Pomerleau 1988; Bain and Sammut 1995). At the start of round r it is rolled out on a fresh pool drawn from the task’s reset distribution, and those episodes it fails form the round’s failure set,

$$\mathcal{F}_r = \{ f_i = (\tau_i, \ell_{1:T_i}^{(i)}) : \tau_i \text{ is a failed rollout of } f_\theta \}, \quad (2)$$

with $N = |\mathcal{F}_r|$, which varies by round, and T_i the length of trajectory τ_i . The round closes by acquiring D demonstrations from the expert π^* and aggregating them (Ross,

Gordon, and Bagnell 2011),

$$\begin{aligned} \mathcal{D}_r &= \mathcal{D}_{r-1} \cup \{d_{r,1}, \dots, d_{r,D}\}, \\ \theta &\leftarrow \arg \min_{\theta} \mathbb{E}_{(s,a) \sim \mathcal{D}_r} [\mathcal{L}_{\text{BC}}(f_{\theta}, s, a)] \quad \text{for } r \equiv 0 \pmod{m}, \end{aligned} \quad (3)$$

halting when $|\mathcal{D}_r| - |\mathcal{D}_0| = B$, so the loop runs B/D rounds. Retraining runs at a per-task cadence m set identically for every arm. Success is read on a frozen held-out set that every arm shares and no arm sees during acquisition.

Every gated method compared here instantiates the same three-part acquisition rule $A = (t^*, C_{\text{tgt}}, \xi)$, where t^* is the step at which the expert takes over (when), C_{tgt} the failure mode the round is spent on (which), and ξ the state from which the corrective demonstration begins (where). The query gates compute t^* and inherit the other two from the rollout that tripped the threshold; the uniform-random control fixes all three by sampling. DISEIL adopts the same t^* and computes the remaining two, changing nothing else in the loop, so the comparison is controlled.

DISEIL

DISEIL runs four stages once per round (Figure 2): perceive the round’s failures, partition them into failure modes, prioritise one mode, and prescribe the D demonstrations the expert is asked to supply. Two models fill three roles. The *vision-language model* (Bai et al. 2025) reads rendered frames. The *prescription model* (PM) (Yang et al. 2025) is a text-only reasoning model that labels each failure’s root cause and prescribes the round’s request; neither enters the partition.

Perceive. A round opens by describing each failure twice from the same step of the same episode: geometrically, for the partition, and in language, for the prescription. The two never mix. Each failure is anchored at the first step whose loss crosses an out-of-distribution threshold η and stays across it for K consecutive steps,

$$t_i^* = \min \{ t : \ell_u^{(i)} > \eta \quad \forall u \in [t, t + K - 1] \}, \quad (4)$$

falling back to $\arg \max_t \ell_t^{(i)}$ if no crossing occurs. The threshold is a quantile of the training-loss distribution, recalibrated at every retrain; the construction is Diff-DAGger’s, used unchanged (Lee, Kang, and Kuo 2025). The peak arrives late, so an expert taking over there inherits a corrupted state with little episode left.

Each failure is then reduced to a six-dimensional vector computed from the privileged simulator state at t_i^* , on the manipulation tasks

$$\begin{aligned} \phi_i &= [p_{x,i}, p_{y,i}, \sin \psi_i, \cos \psi_i, \rho_i, \delta_i], \\ \rho_i &= t_i^*/T_i, \quad \delta_i = \|p_i^{\text{ee}} - p_i^{\text{obj}}\|_2, \end{aligned} \quad (5)$$

where $(p_{x,i}, p_{y,i})$ is the planar position of the task-relevant object and ψ_i its yaw, entered through its sine and cosine so the wrap at $\pm\pi$ does not create a false distance, ρ_i the fraction of the episode completed before the flag, and δ_i the end-effector-to-object distance. The same six slots are refilled per task. The descriptor is small because the failure sets are small: over a few dozen points, weakly informative

dimensions flatten the pairwise distances the merge order depends on, which the supplementary measures.

In parallel, three rendered frames, at the episode’s start, at t_i^* and at its end, go to the vision-language model, which returns a short spatial account. The prescription model converts it into a root cause and a trajectory phase drawn from a closed taxonomy stored in $\mathcal{K}$ rather than invented. The division of labour follows the evidence: such models name a cause competently when given structured evidence (Duan et al. 2025; Liu, Bahety, and Song 2023) and reason unreliably about metric geometry from pixels alone (Chen et al. 2024a; Fu et al. 2024).

Partition. The round’s failures are partitioned by a generic clustering step $\mathcal{C}$ over the standardised descriptors,

$$\begin{aligned} \tilde{X}_i &= \frac{\phi_i - \mu_{\phi}}{\sigma_{\phi}}, \quad \{C_1, \dots, C_{k^*}\} = \mathcal{C}(\tilde{X}, k^*), \\ k^* &= \arg \max_{k \in [2, k_{\text{max}}]} \text{sil}(k), \end{aligned} \quad (6)$$

where μ_{ϕ} and σ_{ϕ} are the per-dimension mean and standard deviation of the round’s own descriptors $\{\phi_i\}_{i=1}^N$, and $k_{\text{max}} = \max(2, \min(6, N - 1))$. The step is generic by design: agglomerative clustering is the instantiation used here (Ward 1963), k-means would serve (Lloyd 1982), and k^* is chosen by the silhouette criterion, used unmodified (Rousseeuw 1987; Pedregosa et al. 2011), so DISEIL claims a partition step and not its implementation. Each mode carries its raw-pose centroid, its mean peak loss $\bar{L}_C$, the mean over its members of $\text{peak}_i = \max_t \ell_t^{(i)}$, and a representative $\text{rep}(C)$, the member nearest the cluster mean in standardised space; the dominant mode C^* has the most members, ties broken by mean peak loss. The peak scores a failure while the first crossing locates it. Below four remaining failures each failure becomes a singleton and Equation 7 reduces to selecting the highest-peak-loss failure. A mode is named by the majority root cause among its members, since a method that reports a failure in mode 2 has said nothing.

Prioritise. This stage makes the first of the two decisions DISEIL adds, the which; Prescribe below makes the where. The round is spent on the mode of highest mean peak loss among those within one member of the largest,

$$C_{\text{tgt}} = \arg \max_{C: |C| \geq |C^*| - 1} \bar{L}_C. \quad (7)$$

The size constraint keeps the target inside the bulk of the round’s failures, so a mode that barely exists cannot capture the round’s budget on the strength of one badly failed episode. Because one demonstration rarely removes a mode outright, a rule of this shape can return the same mode round after round. DISEIL therefore has a cluster memory, a configurable task-specific component outside the core rule, which discounts modes already corrected and is on in every run here.

The prescription model is shown at most κ cited failures from the target mode: the representative first, then the worst-loss failure, then farthest-point selection (Eldar et al. 1997) to span the mode.

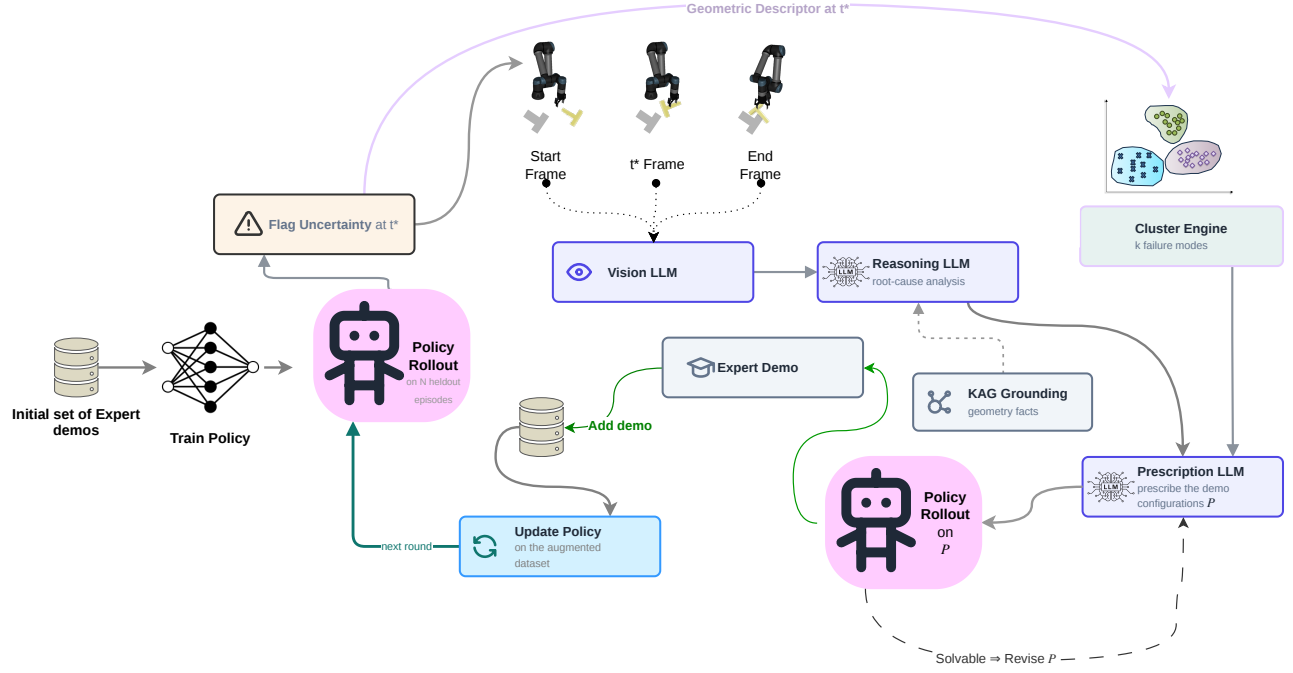


Figure 2: DISEIL. The loss flag (Eq. 4) returns the step t^* at which the policy first becomes unreliable, which is the whole of what a query gate computes and the point at which DISEIL continues. The perception branch meets the geometric branch only at the prescription, so the partition consumes no model output. “Reasoning LLM” and “Prescription LLM” are the two roles of the prescription model, a large language model; “KAG Grounding” is the knowledge-augmented-generation constraint graph $\mathcal{K}$. The solvability screen, drawn as the rollout and revise branch, is specified in the supplementary.

Prescribe. The prescription model receives the target mode’s anchor geometry, the cited failures with their root-cause labels, and the task’s constraints, and returns the round’s request with a confidence score and a rationale. Each requested demonstration takes one of two forms. A *targeted correction* names one cited failure; that exact episode is re-instantiated and the expert takes over at the flagged step. A *bridging placement* names two or three cited failures and asks for a new configuration between them, from which the expert demonstrates a complete episode. Bridging lets a prescription be easier than any failure it addresses: where a mode lies far outside what the policy can solve, a targeted correction is a large distributional jump and a bridged one is a step the policy can absorb. The graded-placement argument is reverse curriculum generation’s (Florensa et al. 2017), applied to a demonstration request rather than a training reset.

A model asked for a configuration of the world will sometimes name one the world cannot produce. The graph makes such a request checkable, holding workspace bounds, spawn ranges, reachability, controller limits, the success predicate and the failure-mode vocabulary as structured knowledge, in the manner of a robot knowledge base (Tenorth and Beetz 2013) rather than a retrieval index over text (Lewis et al. 2020; Edge et al. 2024). Verification is a loop: the model proposes, a map g turns the proposal into a reset specification, the specification is checked, and a violation is returned

as feedback so the model can propose again,

$$\begin{aligned}
 \text{cmd}^{(j)} &= \text{PM}(C_{\text{tgt}}, S, \mathcal{K}, \text{violation}(\xi^{(j-1)})), \\
 \xi^{(j)} &= g(\text{cmd}^{(j)}), \\
 V(\xi) &= \mathbf{1}[\xi \in \mathcal{W}_{\mathcal{K}}] \wedge \mathbf{1}[\text{reachable}_{\mathcal{K}}(\xi)] \\
 &\quad \wedge \mathbf{1}[\text{valid-path}_{\mathcal{K}}(\xi)],
 \end{aligned} \tag{8}$$

where $\mathcal{W}_{\mathcal{K}}$ is the workspace bounds, the conjuncts of V are the constraints $\mathcal{K}$ stores for that task, and $\text{violation}(\xi^{(0)}) = \emptyset$ on the first attempt; on the grid task the last conjunct is a path-validity predicate decided by A* and breadth-first search (Hart, Nilsson, and Raphael 1968; Cormen et al. 2022), which are never the expert. A failed attempt consumes no budget, since the budget counts demonstrations collected and not prescriptions proposed, and after J_{max} attempts the round falls back to the nearest untried recorded failure. A second screen, which rejects a prescription the policy already solves, is specified in the supplementary and is not exercised in any run below.

Every stage bears on what the policy is trained on and none on how it is fitted, so the evaluation varies the acquisition rule and holds the rest of the loop fixed. The supplementary states the round as pseudocode.

Task	Obs	$ \mathcal{D}_0 $	Init SR	Safe	Dropout	Ensemble	Thrifty	Stagger	Diff-DAGger	DISEIL (ours)
GridWorld	state	20	48.9	85.3 $\pm$ 0.9	84.9 $\pm$ 0.8	86.2 $\pm$ 0.7	86.8 $\pm$ 0.7	85.7 $\pm$ 0.5	–	92.4$\pm$0.4
GridWorld	image	20	47.0	88.8 $\pm$ 0.9	88.4 $\pm$ 0.7	88.8 $\pm$ 0.9	88.7 $\pm$ 0.6	89.1 $\pm$ 0.8	–	91.3$\pm$0.6
Push-T	state	20	46.2	82.0 $\pm$ 3.0	84.8 $\pm$ 2.7	85.9 $\pm$ 2.6	83.2 $\pm$ 3.2	–	94.1 $\pm$ 2.0	96.1$\pm$1.6
Push-T	image	20	43.3	78.1 $\pm$ 3.5	82.1 $\pm$ 3.1	83.2 $\pm$ 3.0	79.3 $\pm$ 3.6	–	89.0 $\pm$ 2.1	92.6$\pm$2.2
Lift	state	8	67.2	99.2 $\pm$ 0.7	99.2 $\pm$ 0.4	99.2 $\pm$ 0.4	100.0$\pm$0.0	–	99.2 $\pm$ 0.4	100.0$\pm$0.0
Lift	image	8	66.4	99.6 $\pm$ 0.4	97.2 $\pm$ 1.6	98.8 $\pm$ 0.7	99.6 $\pm$ 0.4	–	99.6 $\pm$ 0.4	100.0$\pm$0.0
Wipe	state	12	47.7	88.0 $\pm$ 1.1	88.6 $\pm$ 1.8	86.8 $\pm$ 1.9	89.0 $\pm$ 1.1	–	90.4 $\pm$ 2.7	93.1$\pm$1.3
Wipe	image	12	45.2	69.6 $\pm$ 2.4	83.2 $\pm$ 3.0	84.4 $\pm$ 3.2	69.2 $\pm$ 4.0	–	88.6 $\pm$ 1.4	92.3$\pm$1.4
Door	state	4	56.8	91.8 $\pm$ 2.1	92.5 $\pm$ 1.2	88.8 $\pm$ 3.1	89.6 $\pm$ 1.7	–	93.2 $\pm$ 1.9	96.6$\pm$1.9
Door	image	4	43.1	82.4 $\pm$ 1.4	81.8 $\pm$ 1.5	83.0 $\pm$ 4.9	82.8 $\pm$ 1.2	–	84.2 $\pm$ 1.6	88.6$\pm$1.5

Table 1: Final held-out success rate (per cent), mean $\pm$ standard error over 5 seeds (robot tasks) and 9 seeds (GridWorld), at a budget of twenty expert demonstrations. $|\mathcal{D}_0|$ is the number of initial demonstrations, Init SR the round-0 success rate, and the best mean in each row is in bold, ties bolded jointly. Stagger is a uniform-random allocation control; the rest are query gates. A dash marks a method that does not apply: Diff-DAGger on GridWorld, whose policy is not a diffusion policy, and Stagger where it was not run. The Lift rows are at the ceiling, and the tie they show is not a tie in sample efficiency: DISEIL reaches 100 per cent after the ninth demonstration on Lift (state) and the seventeenth on Lift (image), ThriftyDAGger only after the seventeenth on Lift (state).

Task	Obs	Diff-DAGger	DISEIL
Push-T	state	1.57 $\pm$ 0.49	2.81$\pm$0.93
Push-T	image	1.80 $\pm$ 0.49	2.82$\pm$0.77
Lift	state	1.61 $\pm$ 0.50	2.64$\pm$0.74
Lift	image	1.36 $\pm$ 0.38	2.93$\pm$0.75
Wipe	state	1.43 $\pm$ 0.36	2.91$\pm$0.90
Wipe	image	1.95 $\pm$ 0.52	3.62$\pm$0.98
Door	state	1.84 $\pm$ 0.50	3.43$\pm$0.95
Door	image	1.58 $\pm$ 0.41	3.00$\pm$0.89

Table 2: Per-demonstration information gain, mean $\pm$ standard error over the same seeds: the policy’s per-step loss on each newly acquired demonstration, measured before retraining on it, with the higher value in bold. Diff-DAGger is the one baseline whose gate selects on the same per-step loss the metric is computed from, and it is lower in all eight settings in which it runs. GridWorld is omitted because Diff-DAGger does not run there.

Experiments

Setup

Tasks and policies. We evaluate on five tasks under two observation modalities, state and image, for ten settings in total. GridWorld is a 5×5 discrete grid with three obstacles. Push-T is the ManiSkill3 `PushT-v1` environment (Tao et al. 2024) of the task introduced by Florence et al. (2021); Lift, Wipe and Door are RoboSuite UR5/UR5e tasks (Zhu et al. 2020). The policy class is fixed by the modality and shared by every arm: an MLP on GridWorld (state), a CNN on GridWorld (image), and a diffusion policy (Chi et al. 2023) on the four robot tasks, with an R3M encoder (Nair et al. 2022) on the image branch. R3M supplies the policy’s features and never enters the partition, which is geometric in every run. The expert is a human on GridWorld and a scripted or motion-planned oracle on Lift, Door and Wipe. On Push-T it is a PPO policy (Schulman et al. 2017), competent in one

rotational direction only, and the task’s graph excludes the configurations it cannot serve. That graph encodes the expert’s competence region, which the gates cannot represent, so on Push-T part of DISEIL’s margin may be avoidance of expert failure rather than better allocation.

Budget and measurement. Every arm receives $B = 20$ expert demonstrations at $D = 1$ per round and retrains on a shared cadence m , from scratch every round on GridWorld ($m = 1$) and on every fourth acquired demonstration on the robot tasks ($m = 4$). Three rounds in four therefore analyse a policy that has not been refitted. Each still draws a fresh rollout pool, so the failure set and its partition are resampled rather than repeated. We report final held-out success rate as mean $\pm$ standard error, with $\text{SE} = \text{std}/\sqrt{n}$ over $n = 9$ seeds on GridWorld and $n = 5$ on the robot tasks. The initial set $|\mathcal{D}_0|$ is set per setting by a behaviour-cloning data-scaling sweep targeting a round-zero success rate near 50 per cent, reported as Init SR in Table 1. That band is where a fixed budget can be spent well or badly: below it the failure set carries no structure, above it it is empty. The ten settings are not ten independent experiments. The two modalities of a task share the expert, the reward structure and the reset distribution, so the ten rows carry less evidence than ten separate tasks would.

Baselines. The comparison is against the query-efficient members of the DAGger family (Ross, Gordon, and Bagnell 2011): SafeDAGger (Zhang and Cho 2017), DropoutDAGger (Menda, Driggs-Campbell, and Kochenderfer 2017), EnsembleDAGger (Menda, Driggs-Campbell, and Kochenderfer 2019), ThriftyDAGger (Hoque et al. 2021a) and Diff-DAGger (Lee, Kang, and Kuo 2025). Each instantiates the gate template of the Related Work, so they differ in the scalar and not in what the round’s demonstration is spent on. Diff-DAGger’s scalar is the per-step denoising loss, the same signal DISEIL uses to localise a failure; it runs on the robot tasks only. We add Stagger, a uniform-random allocation control that corrects one uniformly chosen recorded failure, with no gate of any kind. Running it on all eight robot settings was

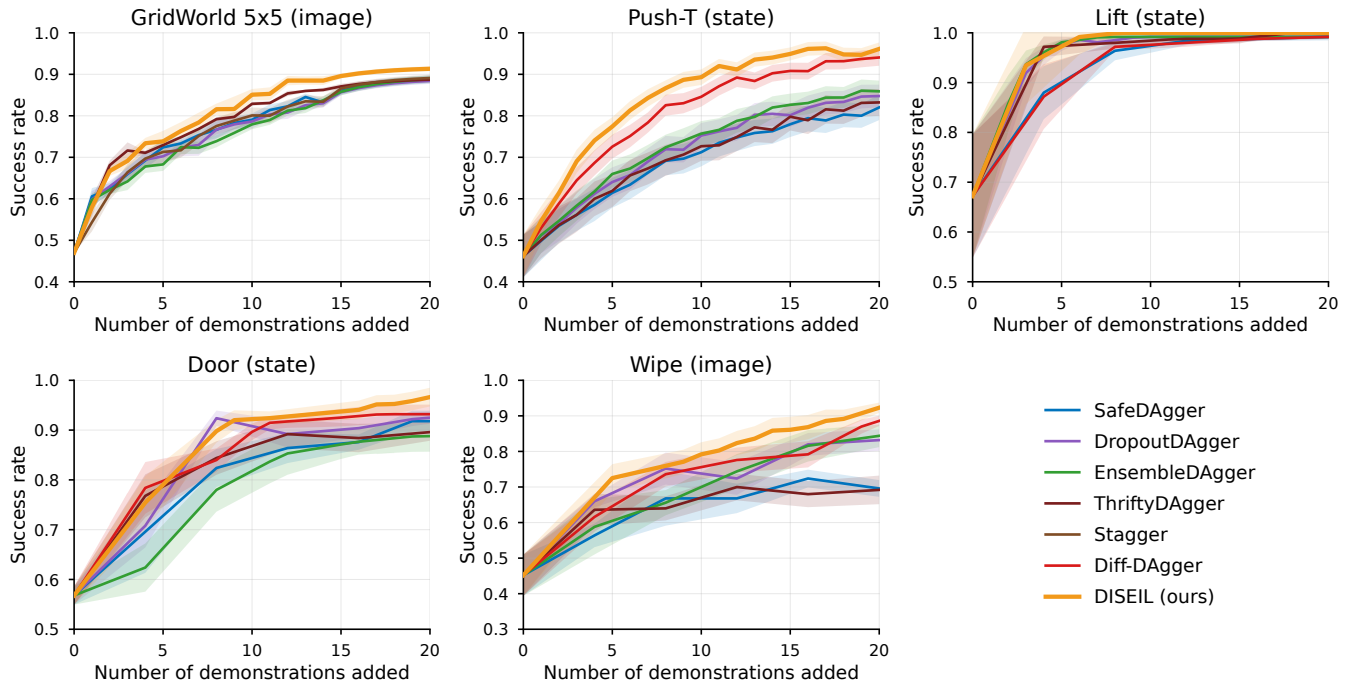


Figure 3: Held-out success rate against demonstrations acquired, mean $\pm$ standard error, on five settings spanning a discrete CNN policy and a continuous diffusion policy. On Push-T the separation opens by the fifth demonstration and holds; on Wipe (image) both arms are still rising at the budget.

beyond the compute available, so it exists on both GridWorld settings and, as ablation A2 below, on two robot settings. One asymmetry favours DISEIL: its descriptor is read from privileged simulator state, whereas every gate scores the policy’s own output. The descriptor supplies only the partition, and no arm’s policy observes it.

Main results

Best mean in all ten settings. Table 1 gives the final held-out success rate of every arm. The margin over the strongest baseline in each setting averages 2.80 points, with a standard deviation of 1.73 and a range from 0.0 to +5.6. The baseline to be beaten moves between rows: Diff-Dagger on Push-T, Wipe and Door under both modalities, ThriftyDagger on GridWorld (state), Stagger on GridWorld (image), a tie with ThriftyDagger on Lift (state), and three baselines level at 99.6 on Lift (image). What carries the result is the recurrence of the margin across the rows rather than the size of any one of them: the seed standard errors overlap in several rows, and no row would support the claim on its own. On GridWorld (image) the random control finishes level with every gate, at 89.1 against 88.4 to 88.8, so the gate signal buys nothing there while DISEIL still gains 2.2 points, which is consistent with the ablation that follows.

Lift is at a ceiling, and the ceiling hides the result. DISEIL and ThriftyDagger both reach 100.0 ± 0.0 on Lift (state), so the row reads as a tie. It is not one: DISEIL is at 100 per cent after the ninth demonstration and ThriftyDagger only after the seventeenth, so the same endpoint costs eight fewer

expert calls. On Lift (image) DISEIL reaches 100 per cent after the seventeenth. A final number cannot express that once both arms have run out of headroom, and Lift begins the budget furthest above the target band, at 67.2 and 66.4. Where a framework and its baselines both sit at the ceiling, a null result cannot separate a component that does nothing from one whose effect cannot be observed, so the ablation programme runs only where headroom remains.

Where the margin opens. Figure 3 tracks success rate against demonstrations acquired. On Push-T the separation opens from about the fifth demonstration and holds, so the endpoint in Table 1 reports a gap present for three quarters of the run. On GridWorld every method rises together and finishes bunched, so GridWorld (image) at +2.2 is among the smallest margins. Wipe (image) has not plateaued: both arms are still rising at the twentieth demonstration, so the 3.7-point gap rests on the final values and not on a demonstrated asymptote, and a longer budget could close it.

Information gain

Per-demonstration information gain is the policy’s per-step loss on each newly acquired demonstration, taken before the policy is retrained on it (Table 2). Diff-Dagger is the reference here: its gate selects on the very loss the metric measures, so it is the one baseline that could be expected to lead. It does not. DISEIL is higher in all eight settings in which Diff-Dagger runs, higher than every other gate in all ten, and higher than the control in the four settings in which it runs (supplementary). DISEIL’s own target rule (Eq. 7) also se-

lects on peak loss, so the metric is not neutral to it either, and the table establishes only that the acquired demonstrations are not less informative. A high pre-retrain loss admits two readings, novel data or an incoherent demonstration, and the second is excluded by construction: an infeasible configuration never reaches the expert (Eq. 8), and the demonstration comes from the expert whose trajectories are the fitting target. Two readings survive that Table 1 cannot separate: the demonstrations are better, or the budget is better spread across the failures the policy is still producing.

Ablations

The knockouts settle it. The budget is better spread. Ablations run in three settings spanning a CNN policy, a state diffusion policy and an image diffusion policy, in the order GridWorld (image), Push-T (state), Door (image). Lift is excluded for the reason given above: at the ceiling a null result is unreadable. We report the three values and their signs. The supplementary gives the full programme with margin retained per setting.

Removing the partition costs success without costing information. Deleting the mode structure but keeping the loss signal, so each round greedily corrects the highest-loss failure, is the most damaging knockout, costing $-2.2/ -4.1/ -6.8$ points, a mean of 4.37, and a mean of -53.2 per cent of the margin retained, beneath its own best baseline on Push-T (92.0 against 94.1) and Door (81.8 against 84.2). Per-demonstration information gain does not fall with it ($+0.02/ +0.16/ -0.13$, mean $+0.02$). Greedy worst-loss selection acquires demonstrations that are individually informative and jointly redundant, so the partition buys coverage of the failure distribution and not a better demonstration. The knockout removes the whole allocation stack, since the descriptor and the memory have nothing to operate on once modes are gone.

The advantage lies in the allocation. Uniform-random allocation over the recorded failures (A2) reaches $89.1/82.3/80.0$ against DISEIL’s $91.3/96.1/88.6$, below the strongest gated baseline on both robot settings, and promoting the deterministic fallback to the whole method costs 3.27 points, so neither replaying failures nor the heuristic accounts for the result. That fallback knockout keeps the partition and reverts only the placement, the nearest isolation of *where* available here, but it also removes the prescription model, so the two are not separated. Removing the constraint graph costs 2.37 points and drives the fallback rate to 27 to 35 per cent of rounds, and bridging placement costs 1.27; the prescription model, the vision-language model and the cluster memory cost 1.33, 1.33 and 0.73, gaps comparable to the seed standard error of the full run. Replacing the prescription model with a heuristic leaves the Push-T margin at 0.1 points, inside that standard error, so a model-free deployment retains the result on two of the three settings only; dropping the vision-language model as well was not measured. Across budgets the margin is $+9.07$ at $B=10$, $+2.87$ at $B=20$ and $+2.83$ at $B=40$, narrowing because the baselines catch up from a lower start, though DISEIL at $B=10$ does not reach what the strongest baseline attains at $B=20$.

Limitations and Conclusion

DISEIL reasons about one round and holds no representation of the training set. Its only memory records where corrections were placed, not what the data cover, so it cannot tell a failure the set never taught from one it already teaches six times and is merely under-fitted. The descriptor is designed by hand and recovers cause only where configuration determines cause, with root-cause purity per cluster at 0.84 to 0.91 scored against the prescription model’s own labels, so it records agreement inside one system. It is also read from privileged simulator state, so the image results show the allocation transferring to an image policy, not the partition built from pixels; outside simulation it would need a pose estimator whose error would enter the partition. The reasoning stack costs 63 to 1,232 seconds per round more than a query gate, at up to 82,116 tokens, though it runs only at selection time and never in the control loop. No knockout varies *where* with *which* held fixed, so the split of the margin between the two is not measured here. Every expert here is a simulator oracle, which is what lets a budget be counted in demonstrations at all. Outside simulation the budget is a person’s time and demonstrators are not uniformly correct (Belkhale, Cui, and Sadigh 2023; Brown et al. 2019), where the information-gain argument fails, since a high pre-retrain loss identifies novel data only if invalid demonstrations are excluded by construction.

The *which* and the *where* are free only while the budget is not. DISEIL takes both by partitioning a round’s failures into modes over a geometric descriptor and prescribing a feasibility-checked configuration, and it holds the best or joint-best mean success rate in every setting reported here. Removing the allocation stack leaves per-demonstration information gain untouched while success falls by a mean of 4.37 points across the three settings ablated: the demonstrations stayed informative and stopped covering the failure distribution. The components of the *where* each cost between 0.7 and 2.4 points, inside the seed standard error of the full run, so this study does not measure what the prescription is worth on its own. Choice of gate moves the result by as much as 19 points, on Wipe (image), yet no gate exceeds DISEIL in any of the ten settings. What a round buys turns on which failure it corrects, not only on which scalar opens the gate.

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